

Does Renovation Always Pay Off?

A Data Analysis of Housing Prices



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neue fische Data Science Bootcamp

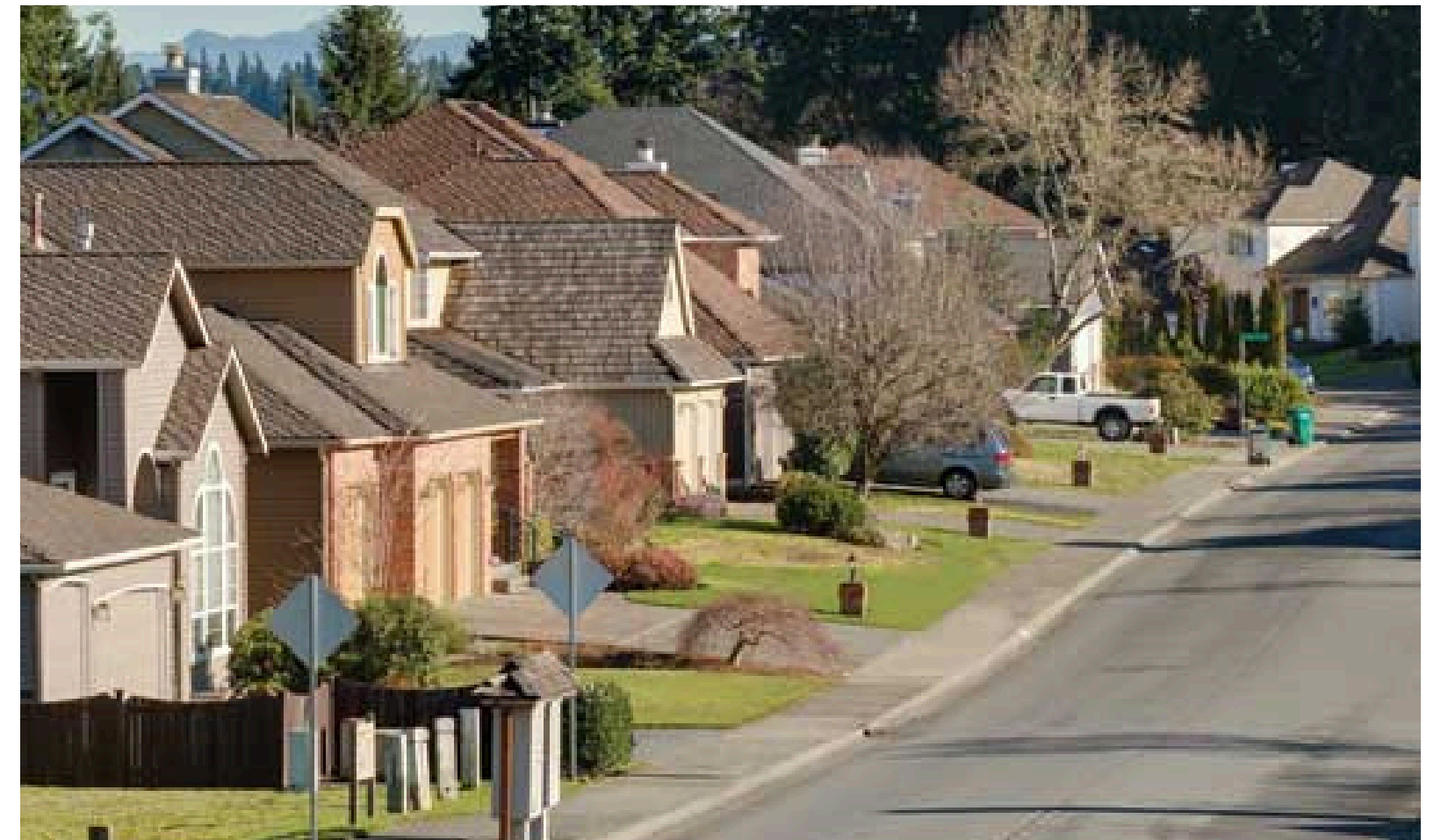
Dec. 2025

A Real Estate Decision: Renovate or Not?

Client: Charles Christensen

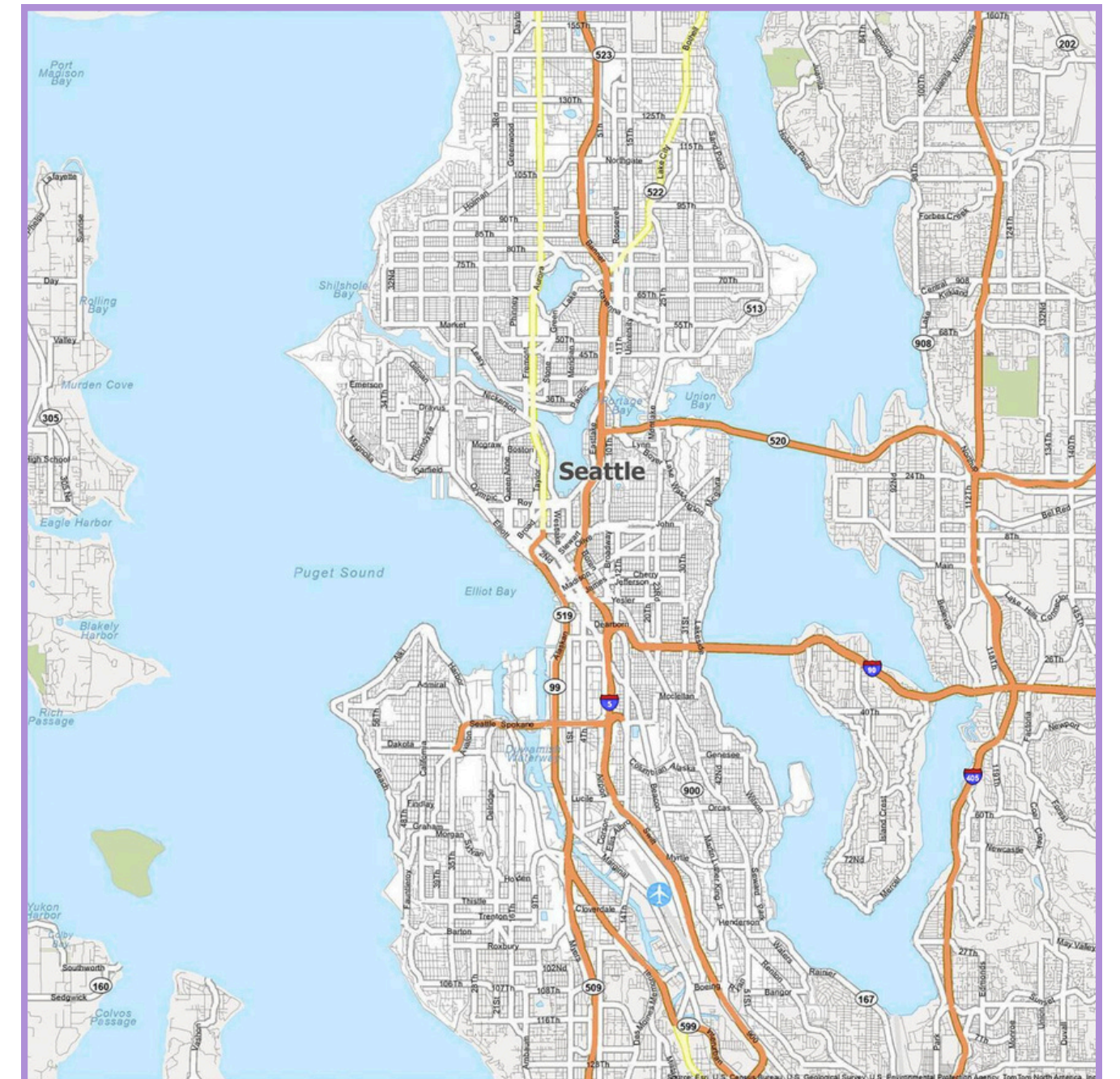
Goal: Maximize return

Key question: Renovate before selling, or not?

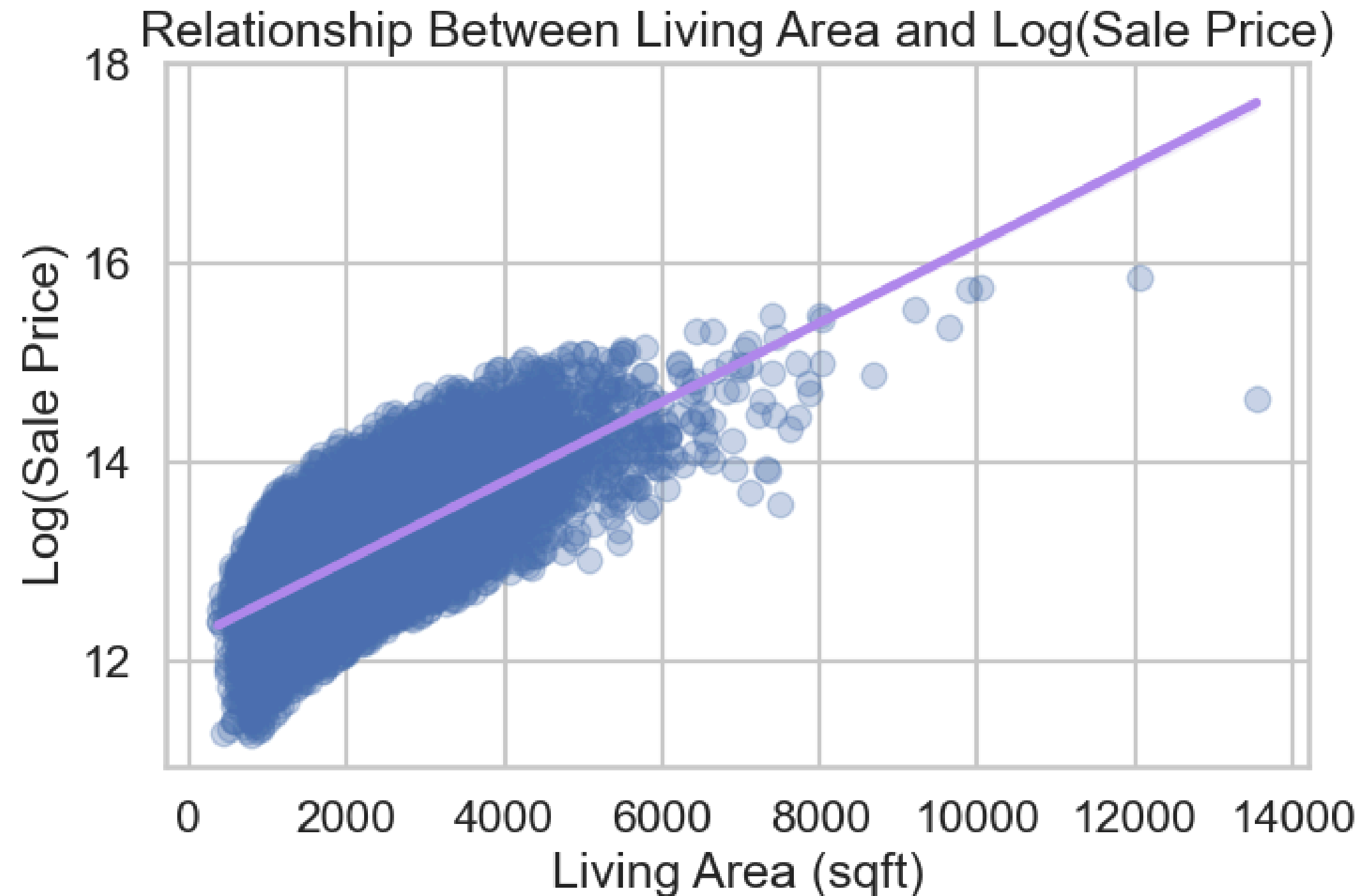


The Data Behind the Analysis

- One year of housing sales data (King County, USA)
- Property size, quality, renovation status, location
- Sale prices and sale dates
- ~21,000 transactions

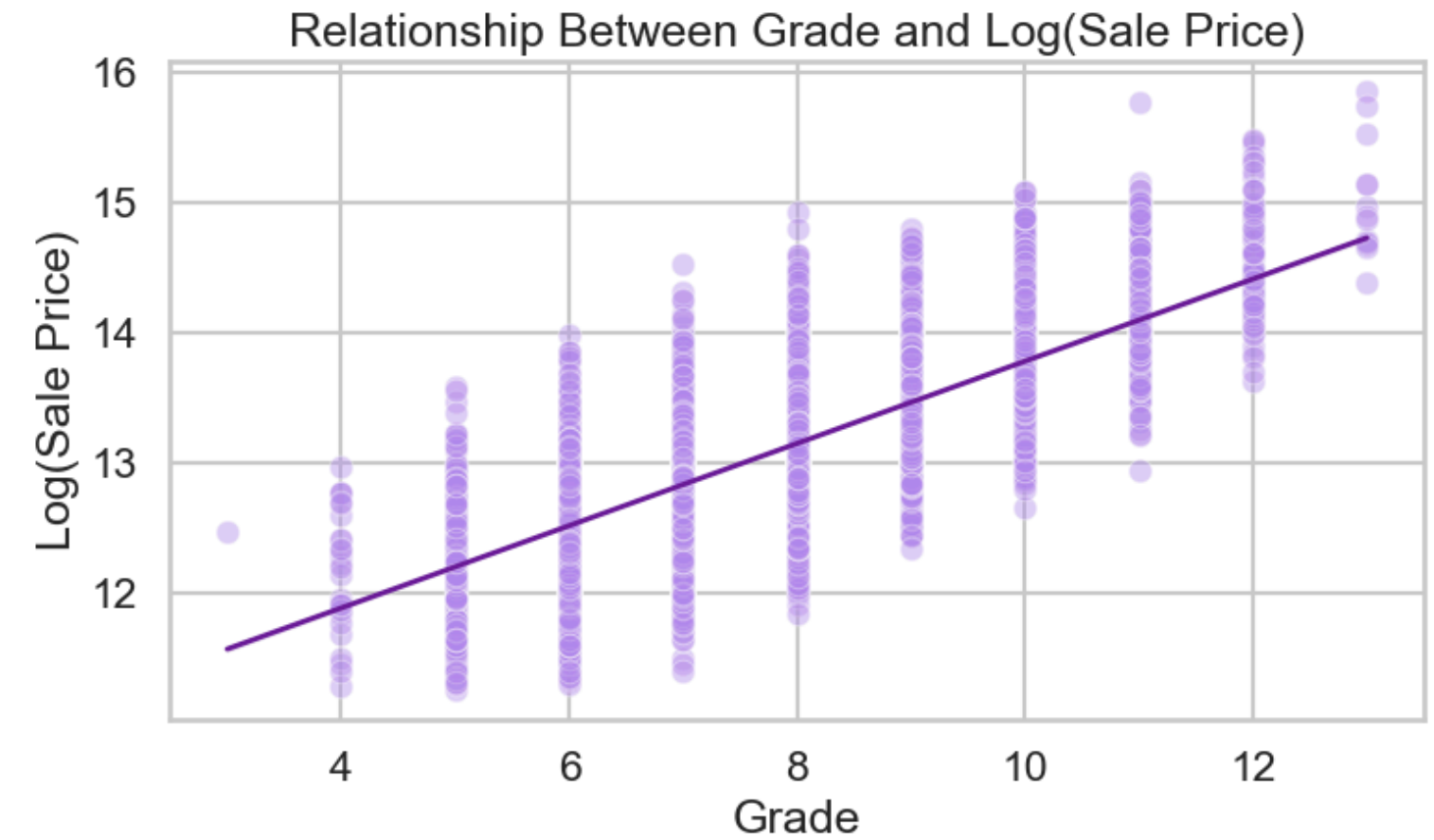
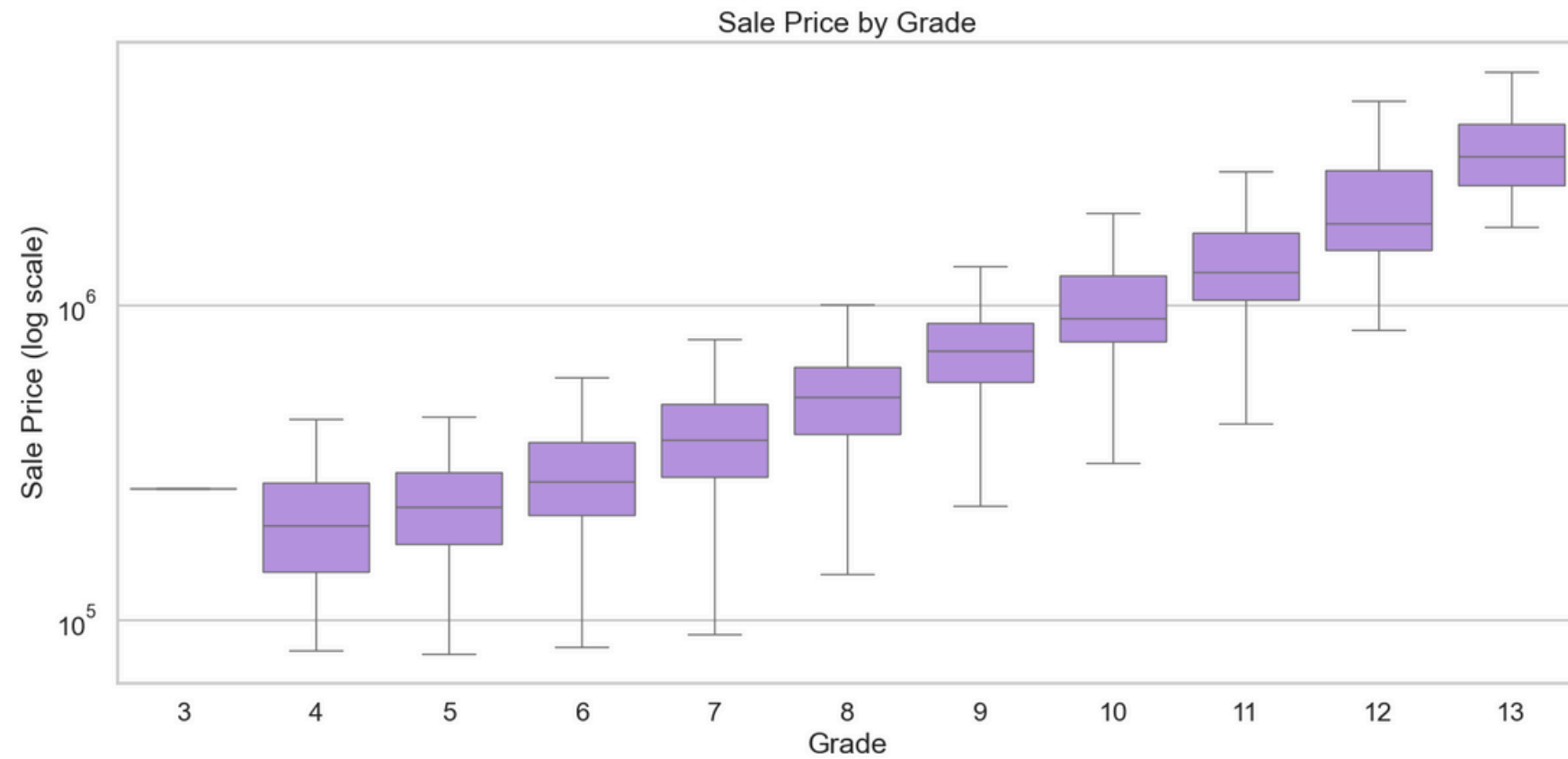


H1: Bigger Houses Sell for Higher Prices



Larger homes generally sell for higher prices

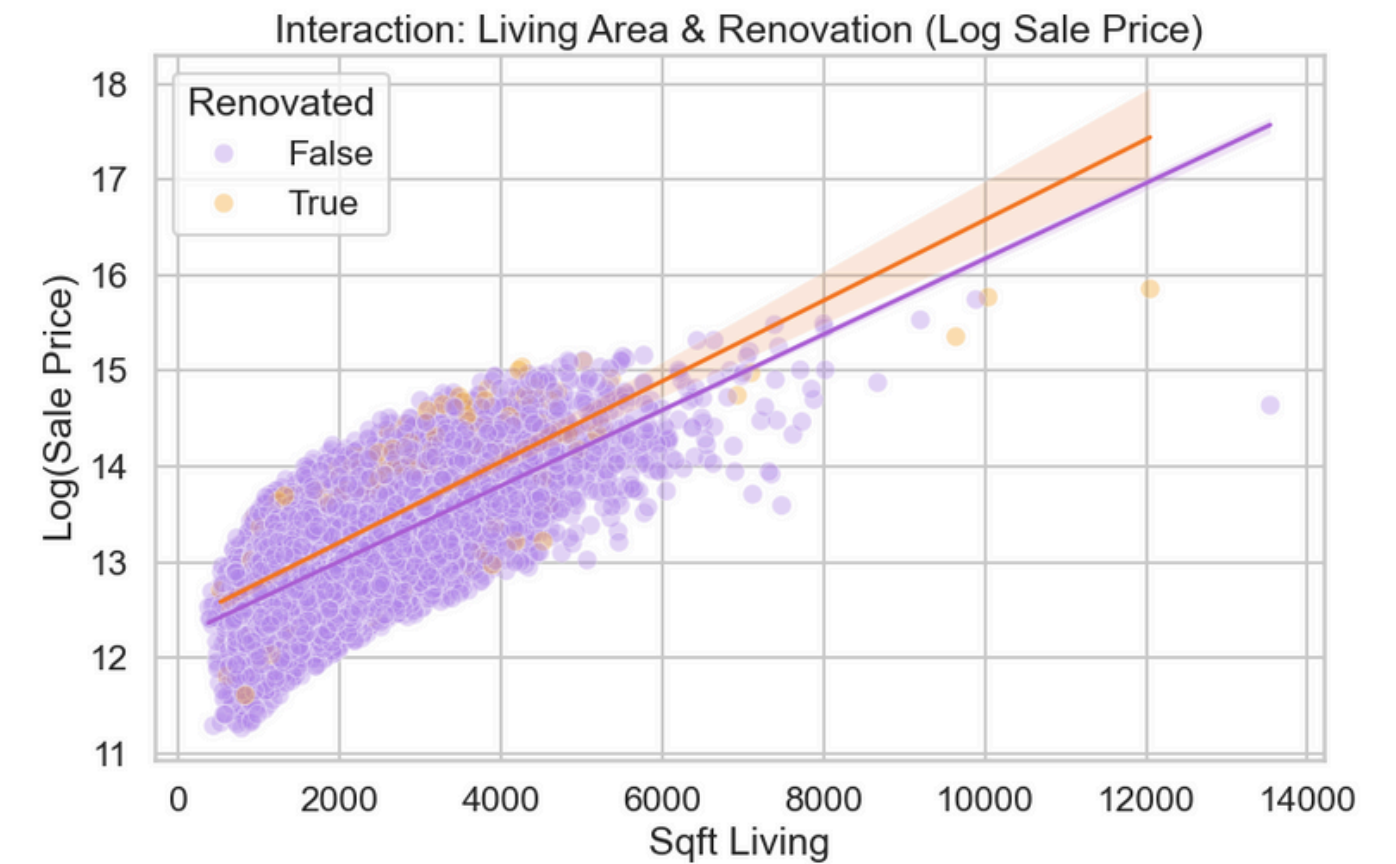
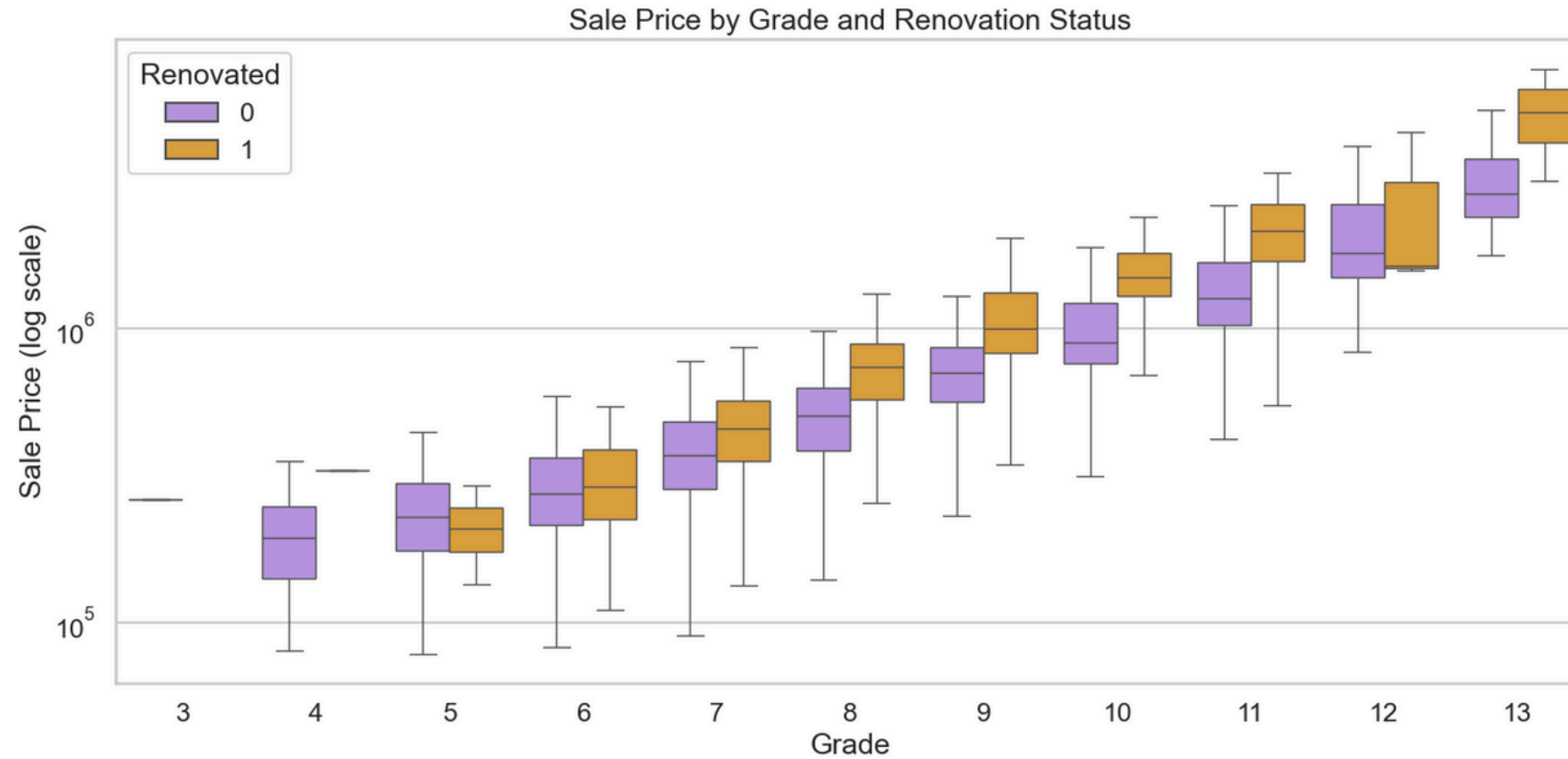
H2: House Quality Has a Strong Impact on Price



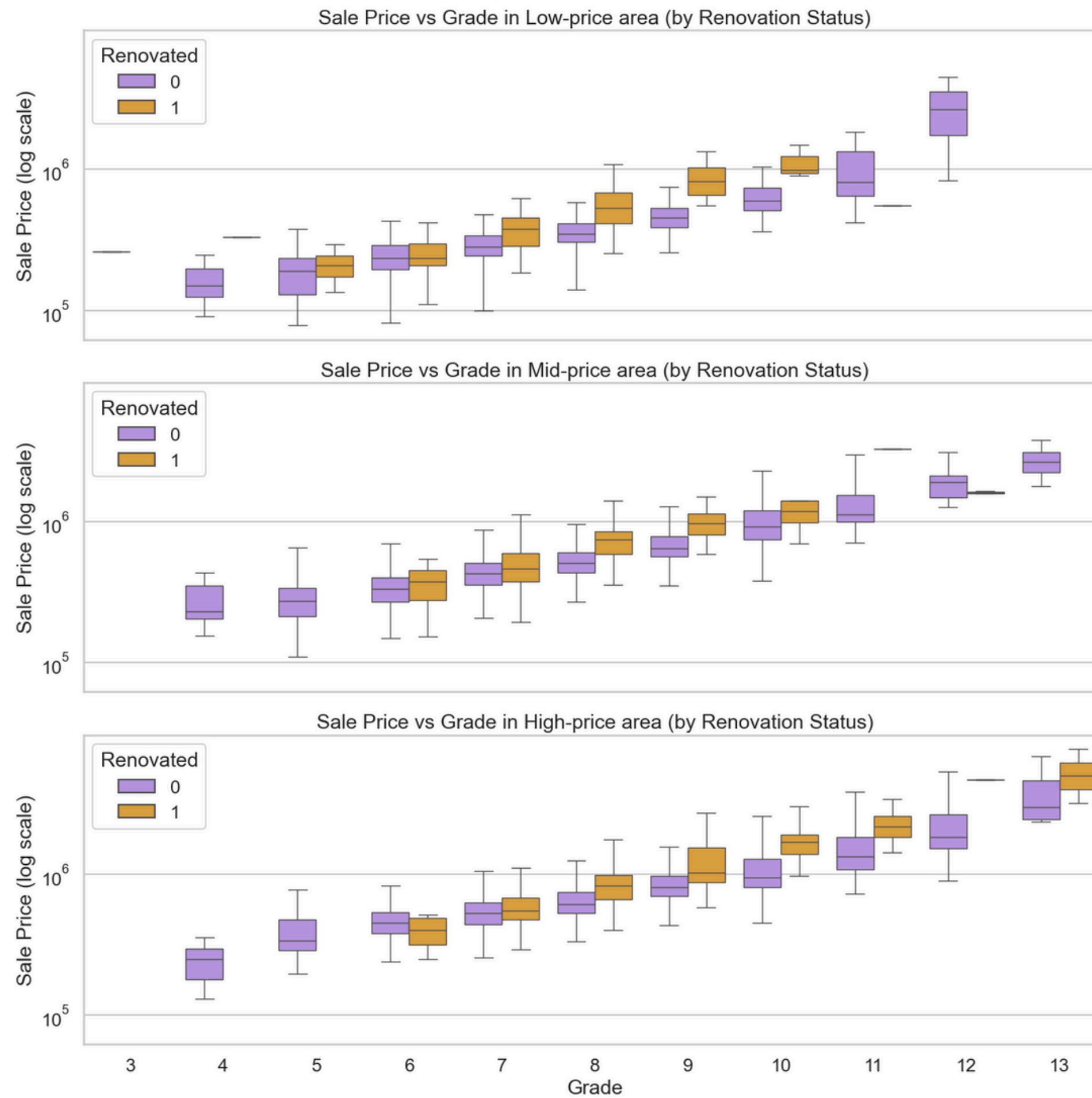
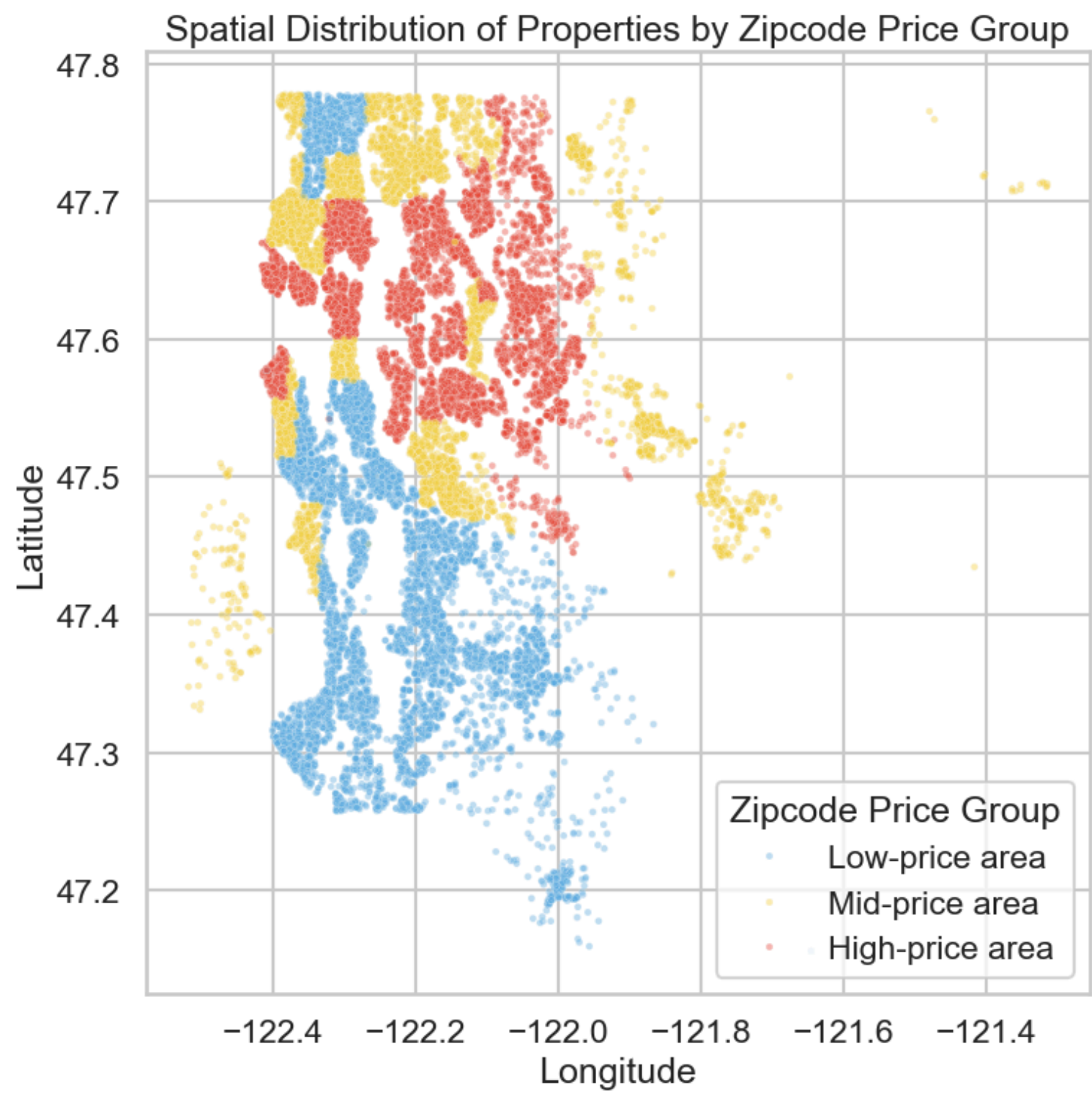
H3: Renovated Houses Sell for Higher Prices



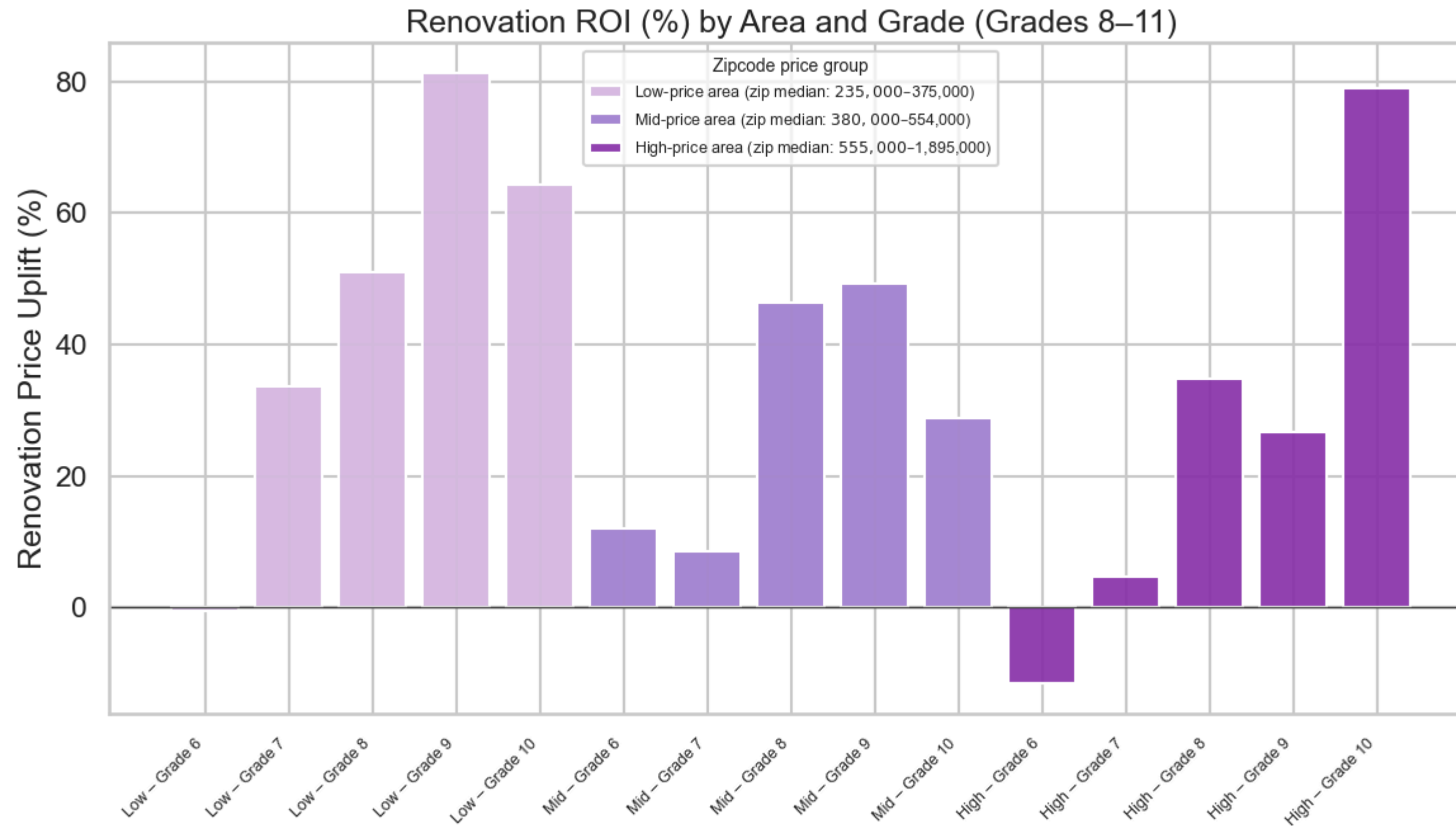
Why One Feature Alone Is Not Enough



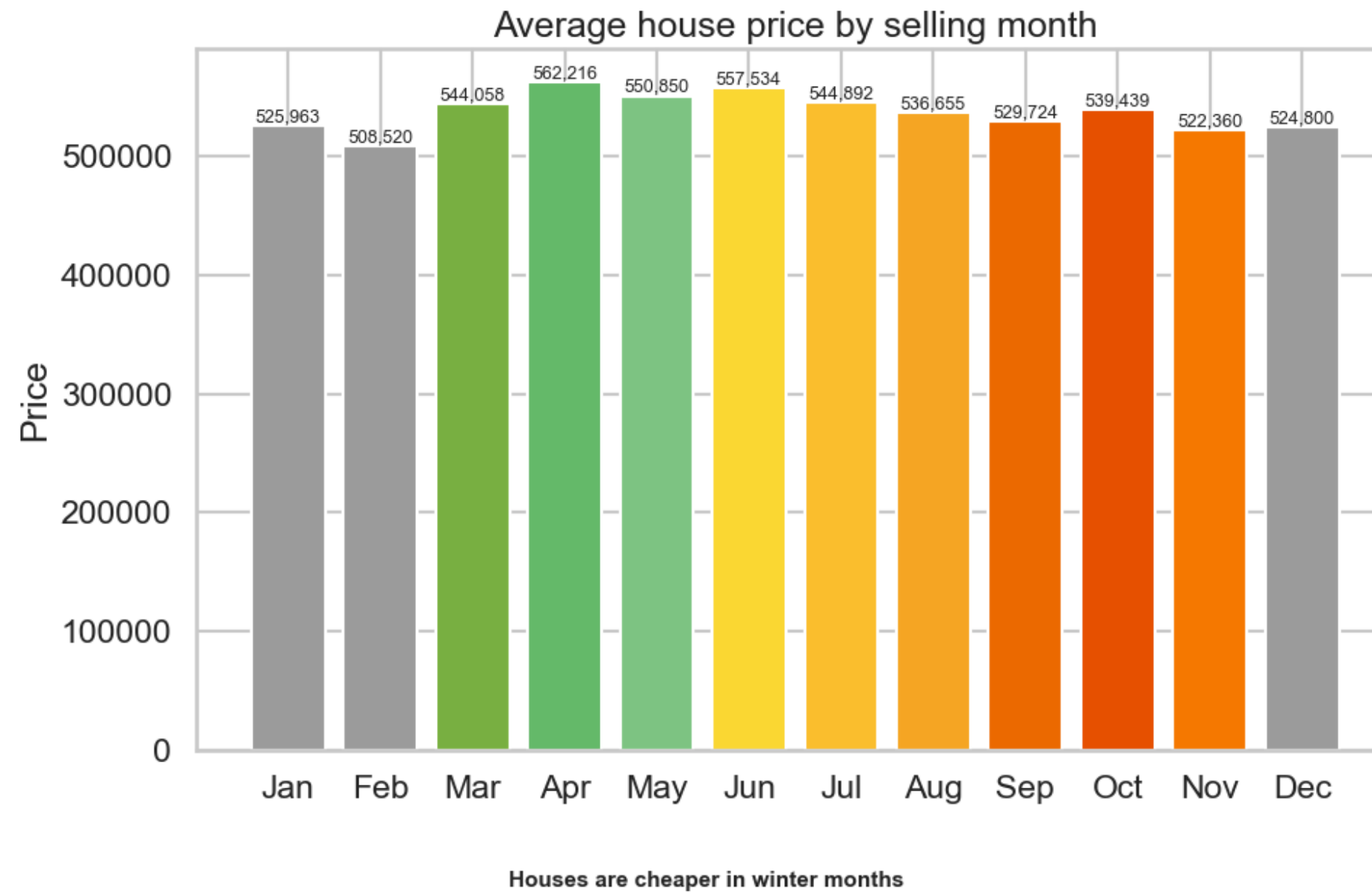
Renovation Depends on Both Location and Quality



Turning Insights into Recommendations for Charles



Seasonal Pattern in Sale Prices (Single-Year Evidence)



So... Should Charles Renovate and When Should He Sell?

Not “always renovate”, but “renovate selectively”

- Focus on houses with grades 8–10
- Avoid very low-grade and very high-grade properties
- Consider the zipcode price level carefully
- Renovate only where the expected uplift is strong
- Aim to sell in spring rather than winter

